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Exploration of health information needs of persons affected by complex cancers and evaluation of co-designed video-assisted health education.

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Abstract

Purpose

Persons with cancer and their families are given a range of complex information (written and verbal) to assist them in managing treatment and side effects at home. This study explored the health information needs of patients and family members and investigated whether video-assisted health education improved understanding of information for persons with complex cancers.

Method

A co-design framework with health consumers and clinicians was used to identify concepts and create videos. Qualitative interviews and thematic analysis of participant demographics explored the health information needs and acceptability of the videos. Persons affected by brain, head and neck and gastrointestinal cancer were included.

Results

Eleven interviews were conducted with patients and family members, aged between 39 to 82 years. The health literacy levels reported by participants highlighted the need for help with medical information and forms. Four themes were developed: sorting through information, acceptability of videos, balance of caring, and information presentation.

Conclusion

Information provided in different formats and tailored to meet the individuals' health literacy levels can reinforce key points from the health professionals and improve health outcomes. Video-assisted health education improves the patients' and families' understanding of information and ability to make informed decisions regarding their cancer treatment and self-care at home. Digital resources may provide a way to improve the understanding of health information, although digital literacy levels influence access, but this requires further investigation.

Background

A cancer diagnosis can cause high stress, and patients and families struggle to understand the information provided. The number of cancer survivors is increasing due to advances in diagnosis and treatment and an aging population [1]. Cancer treatment has become more complex, with more at-home and outpatient care, nurse-led, and GP clinics [2]. Despite these changes, there is still a need for personalised information to help patients and families make informed decisions. Patients and families

require accessible information on symptom management, treatment education, and self-care to improve their quality of life and meet their supportive care needs [3, 4].

For this study, brain, head and neck, and gastrointestinal cancers were grouped and called complex cancers due to their treatment, side effects and survival rates. The survival rate for patients with brain cancer is lower than 55% in the first year, and survivors and their families experience high levels of psychological stress [5]. For persons with head and neck cancer [HNC] and gastrointestinal cancer [GIT], 70–80% of patients survive the first year if treated early, leading to a need to focus on managing the long-term effects of treatment [6].

During cancer treatment and survival, the family is the main source of support and often works with the patient to understand information and follow care advice at home [7]. To ensure that both the patient and family can understand the information provided, health professionals need to deliver information that is accessible and aligns with their health literacy levels [8]. Health literacy, comprising the ability to comprehend and effectively apply health-related information, shapes both patient and family responses to health information. Using a variety of formats to deliver health information can improve the person's ability to understand and enact the information [9].

In Australia, 84% of people use the internet as their primary health information source, with more than 55% using YouTube to understand concepts [10]. However, the credibility and reliability of online information are often unclear [11, 12]. Hospitals and cancer organisations have created websites for cancer information, but much of this information is text-only and lacks videos, making it harder for those with limited health literacy to understand [13]. Health education videos effectively deliver health information [14, 15]. Videos can use narrative or simulation styles to reinforce education on treatment, medication, and physical and psychological changes [16]. Research shows that videos improve patient outcomes compared to verbal education [17, 18]. For example, in India, video education for caregivers of head and neck cancer patients reduced care burdens more than verbal instructions alone, especially for tasks such as medication and dressing changes [17]. Videos also help patients ask informed questions before treatment [14, 16].

The video format enables complex and difficult information to be understood and accessible. Increasingly evidence shows that health information resources should be developed using a co-design framework to ensure resources meet consumer needs [19, 20]. The end user is both the clinician and the health consumer so a collaborative process must involve all end users [21]. The current research extends the research team's published videos and evaluation [15].

The purpose of this study was to explore the health information needs of patients and family members and investigate whether video-assisted health education [VAHE] improves the understanding of persons with complex cancers.

Methods

An exploratory qualitative design underpinned by a co-design framework [19] was used. The co-design phases included identifying needs, concept development and evaluation, using a cyclic process of refining to meet end-user needs. See Fig. 1 Adapted Co-design framework.

To understand health information needs and evaluate the acceptability of video-assisted health education resources the research questions were: ‘What are the health information needs for persons with complex cancers and their families? Does video-assisted health education reduce the unmet needs of people with complex cancers?

Video development

The concepts for each of the videos were decided through discussions with specialist nursing staff at the Cancer Services and three consumer representatives from cancer support groups [brain, head and neck, and women’s cancer groups] to ensure a consumer-led collaborative approach to the development of the concepts [22]. First, a literature review was completed to ensure best clinical practice for the concepts. The next steps involved discussion between the research team and consumers to develop the scripts [14], deciding who would be in the video and the video styles [narrative or simulation]. See Table 1 for video development.

Table 1
Concept to video development

Concept idea	Supporting evidence	Type of video	Equipment	Cultural group
Central access devices- consumer information and consumer fears, trouble shooting at home	(Putranto & Rochmawati, 2020) Clinical Nurse consultant	Narrative with simulation	Device and manikin	No consumer available
Portacath management at home Insertion, management, removal	Clinical Nurse consultant	Consumer story with simulation of management	Device and manikin	Older age group
P**ain management** – Management at home, anticipate pain, nighttime and getting sleep, dealing with nausea, documenting and storage, when to seek help	(Munkombwe et al., 2020)	Narrative with consumer story and nurse information	Simple interview style	2 cultural groups to highlight focus on understanding the consumer
Feeding tube - how to manage at home, when to seek help, what it is and why.	Discussion with Clinical Nurse consultant and medical specialist	Consumer story with simulation of management	Feeding tube and equipment	Patient consumer
Putranto, D., & Rochmawati, E. (2020). Mobile applications for managing symptoms of patients with cancer at home: A scoping review. International journal of nursing practice, 26(4), e12842-n/a. https://doi.org/10.1111/ijn.12842				
Munkombwe, W. M., Petersson, K., & Elgán, C. (2020). Nurses' experiences of providing nonpharmacological pain management in palliative care: A qualitative study. J Clin Nurs, 29(9–10), 1643–1652. https://doi.org/10.1111/jocn.15232				

A vision board was developed for each script with audio and vision sections for filming. This approach enabled concepts to be presented as dialog and key points on screen. Each script included an introduction outlining the aim and a concluding summary by an oncology nurse. The middle part featured a consumer sharing their experience related to the concept presented. This outline meets educational and digital best practice [10, 23]. The script and video development process was shared with consumers and the research team to ensure that the end product was genuinely co-designed. The research team had previously developed and evaluated ten videos [topics including navigating the medical system, mindfulness, sleep, self-care, managing seizures] and the current research developed four videos [managing pain, managing access devices including Portacath, Central access devices and feeding tubes]. The research team's "*Looking Forward After Cancer*" YouTube channel features 23 videos, all accessible to participants from the current study [15].

Study setting and recruitment.

The study was conducted at a metropolitan university hospital in Queensland with a purposive sample of people treated for complex cancer [brain, head and neck, and gastrointestinal cancer] and their family members between December 2022 and December 2023. Inclusion criteria was completing hospital-based treatment and considered low psychosocial risk by the cancer nurse consultants. Exclusion criteria was persons struggling with their diagnosis or treatment due to risk of confusion with treatment information provided in a different format. As per ethics requirements, the clinical nurse specialists for the cancer groups identified potential persons meeting the inclusion and exclusion criteria and provided details to the lead researcher. The lead researcher, EC, contacted the participants to explain the research, obtain voluntary informed consent, provide the video link, and schedule the interview.

Data collection

Individual interviews were conducted by EC an experienced cancer researcher, using Microsoft Teams [24] with instruction given prior to the interviews to ensure understanding of the Teams and the participants were comfortable with this format. After the research was explained and establishing rapport, the interview demographics, health literacy and satisfaction with VAHE were collected. The demographic information consisted of age, gender, patient or family, employment status, and cancer and treatment. BRIEF Health Literacy Screening Tool (BRIEF) is a validated tool to measure health literacy using four questions 5-point Likert, low health literacy is a score below 3/5 [25]. The Satisfaction of Information survey [26] includes five questions with a 5-point Likert scale exploring how satisfied consumers were with the VAHE intervention and care, with five being fully satisfied and one being not satisfied. The interviewer presented a summary to the participants and wrote field notes after each interview. There was no bias with respect to the interviewer as she was external to the hospital. Interview questions were as follows:

- Can you tell me about your health information needs or about your family members information needs?
- Can you tell me about your understanding of how you are managing your health now? For example, managing your diet, staying mobile, symptom management, when to seek help and self-care?
- When reading health information, do you find it difficult, and can you share with me how you work with information provided by the health professional?
- What are their preferences for the delivery of health information?

The study was conducted according to the Australian National Health and Medical Research Council statement on ethical conduct and received approval from the Human Research Ethics Committee Gold Coast Hospital, and the Griffith University Human Research Ethics Committee [HREC/2021/QGC/77927; GU2023/595].

Data analysis

A thematic analysis was used for the qualitative data, and involved identifying, collating, and analysing patterns (themes) within the data [27]. The transcripts were read and reread by members of the researcher team [EC JC NW] to understand the data and to gain a general sense of the interviews. From there, an inductive approach was used to make meaning of the data and categorised into potential themes. The themes were redefined in an ongoing analysis process. Data saturation was achieved at eleven interviews as the research questions focused on information needs during cancer treatment [28]. The mean score of the health literacy survey and the satisfaction of information were collated and reported descriptively.

Results

A total of 21 people were invited to participate in the study and eleven participants recruited and interviews conducted -those with cancer [n = 7] and family members [n = 4]. The reasons for refusal were too busy and too many symptoms and problems. The interviews were between 40 to 60 minutes in length. The ages ranged from 39 years to 82 years. Table 2 shows the participant demographics. The participants' health literacy levels were between three to five, highlighting medium health literacy levels. A specific concept which scored low was needing help understanding the medical information and forms. Satisfaction with the VAHE intervention was rated between four and five points, indicating strong satisfaction, participants identified the videos improved their understanding of the information and they felt respected by the delivery of information. The four themes developed were: sorting through information, acceptability of videos, balance of caring and information presentation. Table 3 shows the qualitative themes and participant quotes.

Table 2
Participant demographics

Code	Role	Age in years	Gender	Cancer diagnosis	Health literacy	Satisfaction 1–5
P1	Patient	75	Male	Transverse Colon Cancer	5	5
P2	Patient	71	Male	Metastatic gastric adenocarcinoma	5	4.8
P3	Patient	82	female	Pancreatic head adenocarcinoma	3.25	5
P4	Patient	49	Male	HNC	4.75	5
P5	Patient	61	Female	HNC	3.5	5
P6	Patient	50	Female	Brain cancer	3.5	4
P7	Patient	31	Female	Brain cancer	2.5	4
C1	Carer	48	Female	Pancreatic head adenocarcinoma	4	5
C2	Carer	39	Female	HNC	5	5
C3	Carer	56	Female	Brain cancer	5	4.8
C4	Carer	58	Female	HNC	4	4.6

P- patient; C- Carer; HNC- Head and Neck Cancer; Health literacy scale 1–5 with five being highest; Satisfaction scale 1–5 with five being highest

Table 3
Qualitative themes and participant quotes

Themes	Quotes from participants
Sifting through information	*You picked up information as time passed and you learned a lot more, it's hard because lots happening at once. My wife captured more because she come with me. After these sessions and some of the stuff we spoke about, I don't recall hearing it, but she had noted it so that was good*. P4 *She's got all the paperwork; she's got every single piece of paper the hospital's ever given her and she's reading it and she keeps misunderstanding stuff.* C1 *Sometimes I find it very difficult because the language is so clinical and there are lots of long words that I don't understand.* P7 *You just pick and grab what you think refers to you at that time* P5 *Be listened to, not like you are overreacting but that you want answers to your questions.* P5
Acceptability of videos	*'Like in everything, you've got to sort of – certainly if you work out what is good and what is garbage basically. Go onto the internet and go through Mayo Clinic and places like that, that have health information* P2 'My *daughter was able to help. Or we're just, Get on the Internet and do a bit of research or ask people, just get advice and information from others'* C3 *I watched and they [research team videos] were reasonably easy to follow and sort of confirmed what I'd understood from the pamphlets* P1 *But a quick video on let's say - when you get your handout there could be like a QR code or something – this is what happens when you go in for chemotherapy.* P2
Balance of caring	*There were times when they [close family] struggled to understand the information and sought help from [extended] family to break the information into options.* C3 '*You got forms to fill out, you've got, appointments to go to. It's a big balancing act and it's very hard to get it right.'* C3 *But yeah, just make sure that you do have that time for yourself. And you do have breaks from friends and support constantly. Friends ringing me. What can I do? How can I help? C3* *I made it a thing I got up every morning. I have two girls, so I committed myself to get up. Didn't matter how I was feeling. I just pushed myself. Yeah, just wanted to get back, I guess to get some normality in it all, I always felt empowered and in control. P2* *Facebook support group was overwhelming for me at the start because I was exposed to everything, all types of cancer and not just the throat cancer. So some of the stories were quite horrific.* P2
Information Presentation	*Well you have to provide all these forms. You have to give them two different doctor forms and they're not like a little note, it's really complex. So it's very hard work.* P6

Themes	Quotes from participants
	Trying to work out words, so, yeah. Yeah. It would be just nice layman terms would be great. C1 "*So if there's a word I don't understand. Yeah. We will google it. I'll say, oh we'll google it. But it means because there's a lot of - there's a lot of big words that you don't understand.* C1 *People are different; there's people that can understand something better by watching rather than by reading. I'm personally one that can do either, either the watching or the reading.* P1 *Talking to others who have gone through the same experience. Pictures with explanations, I am a visual learner and don't take in much at all if it is verbal.* P7 *When you are watching, it's different. You understand more. That's my opinion.* C2 *Yeah, you don't want to put any more pressure on the patient. So it [the information] just has to flow. Everything has to be simple and easy.* C3

Sorting through information

After a diagnosis of cancer, patients are given a large amount of information, all of which have varying levels of importance, and each part of the information is used at different stages of treatment. Working out what information is needed and when was a major aspect of how participants described the information and its use. Participants shared how they gathered and sorted through information to determine what was needed at different times.

Working through the information at home was at different levels with some reading the information and others putting it away for later. Some participants found written information provided by the clinical setting great, while others got lost in the information. Trying to understand the treatment information often led participants to either ask the health professional, but not really understand what they needed to know and then search for answers on the internet. Participants shared how they needed to understand the information enough to work out what they did not know and what search terms to use. Being with the treatment team offered a chance to ask questions but the amount of information often confused participants. '*You ask all these questions, and you forget the main one.'* P3 Finding out the right information that was needed for each phase of treatment and recovery was important for obtaining a level of empowerment, and the ability to develop a plan for managing their life around the cancer treatment. Having a plan helped participants feel in control during the journey. The participants shared how they wanted to be listened to and, have their questions answered as they were working through decisions.

Acceptability of videos

Searching on Google and watching videos for information was something that all participants commented on. Google was the go-to for any words or information they did not understand. The

research team's videos VAHE helped them to understand the tasks that needed to be completed at home. Participants who already had a good understanding of how to navigate digital resources were able to determine what was a good site and what was not reliable. The ability to navigate digital resources was linked to experience rather than health literacy levels.

Searching for information and understanding health information was a family affair and younger family members often helped find the right information. The videos developed by the research team were noted to be easy to watch, accessible and reinforced the information provided by the health professionals and in the pamphlets. The participants explained they watched the video which related to their need for information, for example, the video on ‘managing tube feeding at home’. Although not all participants were able to navigate between the videos within YouTube, this was related to their digital literacy, and how much the participants understood navigating within YouTube.

Balancing information and caring

A diagnosis of a complex cancer is challenging, and participants discussed how they found a balance between caring and sharing information. Information that was presented to the family from the hospital was often misinterpreted when sharing with the family. This caused problems and some families explained how they would ensure that the family was at the medical appointments to fully understand the information. Other families used a style of protective buffering to only share some information with the family. Balancing caring, staying strong, and managing the information were difficult. The family had to manage the treatment schedules and maintain family functioning, for example, keeping their children attending school. Participants who were carers spoke about the need for self-care and the balance of accepting help. Videos on self-care and accepted help, reinforced the need to relax and not think about the cancer. Being about to care for yourself was also empowering and the persons with cancer shared different ways they kept themselves on track. Looking for support outside the family was difficult and participants talked about how they tried different types of support both online and face to face with support groups being sometimes helpful.

Information presentation

The participants all explained that they were provided information as written pamphlets and verbal instructions, however there was a need to tailor the information to their level of understanding. Having the health professional go through the information required for treatment that day rather than explaining the full treatment schedule and expecting them to remember what was expected each week. Completing the medical forms was noted as complex and stressful for both patients and their families. When they could not understand the information, Google was the first place they went to understand the words and information. The participants talked about how the information was presented and how they learn best. This differed across the age groups but having information across different formats was noted to increase understanding. Some participants reported that the information they needed to manage their diagnosis should be easily accessible and presented in simple language. The family members reported that they are balancing caring and managing the home, so the ease of obtaining information was

important. All participants highlighted that they did not want to search through written information to find the part they needed and the research videos VAHE with the visual aspect were noted as helpful when searching for information.

Discussion

This study co-designed and evaluated the acceptability of video-assisted health education and explored the health information needs of persons with complex cancers. The findings showed that the participants were given a great deal of paper-based and verbal information, which was often overwhelming. The thematic analysis identified four themes which were sorting through information, acceptability of videos, balance of caring, and information presentation These themes are visually represented in the model below, illustrating the complex interplay between health literacy, information accessibility, and the identified themes. This visual depiction provides a comprehensive understanding of the study's conceptual framework and practical implications. See Fig. 2: Health Literacy and Information Accessibility Framework for individuals and their families with complex cancers.

The first theme was sorting through information, this theme reflected how patients and families need to read through information to identify what was important at the different stages. Providing a balance of information for patients and families can be challenging for health professionals [29]. The participants talked about different strategies to help them manage the information such as having a family member at the health professional appointment and taking notes. This finding is similar to that of previous research highlighting the need to improve health professional delivery of information to enable the patients and families to identify key points and be able to ask informed questions [30]. One of the influencing factors on a person’s ability to access and understand information is their health literacy levels as this influences their ability to find reliable information [31]. In the current research, the participants shared how they often used Google to fully understand the information they were given.

Linking to health literacy is digital literacy an important factor to consider when developing digital resources [32]. Digital literacy extends the concept of literacy and refers to the ability of the person to access, understand, evaluate, create, and communicate information using digital technologies [33]. This study did not explore or measure this concept. The participants in this study noted that the videos stored within a digital platform YouTube were helpful and reinforced the written or verbal information provided by the doctor and nurses. Similar to previous research, videos were able to provide participants with reliable information in a visual format to enhance their understanding of what to do at home [17]. Some of the participants who reported lower health literacy levels were also found to have less ability to navigate digital information. Meaning that they could watch the video but were unable to navigate the research team’s YouTube Channel to watch specific videos related to what information they needed. This highlights the need for digital resources to be easily provided to patients and families rather than having them search online [12]. Digital literacy is a factor to consider as the health system moves to digital platforms. Patients and families need to be supported to understand and complete information within the digital platforms [34]. The development of information in written formats supported by visual

representation of concepts and links to digital or video information may be a way to enable consumer understanding [35].

The balance of caring theme is related to how information is shared and used within the family. There was a sense from some participants that open sharing enabled a strong supportive family connection, whereas other participants talked about showing a brave face. Past research has shown that families with open communication styles tend to maintain family functioning and sharing even through challenges [3, 36]. Presenting information with pictures enables the reader or viewer to engage and remember key points, which is particularly important when dealing with carer stress and fatigue [35, 37]. The last aspect of the theme, the balance of caring was the need for self-care, several of the videos are about the need for self-care and participants shared how the video reminded them about the need for self-care and the difficulty of balancing caring as both the patient and the family. The burden of cancer is real and research always suggest the need for self-care in both the patient and the family caregiver [3].

The last theme of information presentation was about having information presented in different formats. In the current study, participants reported a high literacy level and were able to source and identify reliable digital resources to supplement the written information they were provided. However, participants in the study with lower health literacy levels often Googled information to understand and could not truly identify whether it was reliable or not and then sought other external advice rather than appearing to not understand the health professional. This is an ongoing concern as cancer care becomes increasingly complex and patients are cared for at home [38].

Using a co-design process when developing resources ensures that the final product meets end-users' needs [21]. In this study, the cyclic co-design process included refilming videos to add captions and shorten the length of some videos based on consumer feedback. The inclusion of patients, families, and cancer nurses on the research team provided insights into consumer informational needs. Previous research confirms that involving consumers throughout the research process ensures that resources effectively meet their needs [14, 39]. When designing educational videos, guidelines must be followed to ensure that health information is evidence-based and presented at the sixth-grade literacy level [10, 11]. Digital health information should focus on education, include signposted concepts, keywords on screen, actionable ideas, and be under five minutes long [12, 41]. Storing resources on YouTube allows for closed captions and language changes, making it accessible to most consumers regardless of socioeconomic status or location. This approach helps address the information needs of rural, remote, culturally, and linguistically diverse, and Indigenous populations [42, 43].

Conclusion

Videos presented as part of patient education provide an authentic educational material allowing patients and families to improve their understanding of the information. This study provides insight into the complexity of the delivery of health information and how patients and families understand this information. Video-assisted health education provides a way to supplement and allow information to be

watched at home, and shared with family to improve the patients and families understanding and ability to make informed decisions regarding their cancer treatment and self-care at home. Information should be provided in different formats and tailored to ensure it meets the person's health literacy levels and reinforces key points to improve health outcomes.

Recommendations

The key recommendation from this research is that health information needs to be tailored to the persons health literacy levels and presented in a range of formats. Providing a video to reinforce health information enables family discussion at home and improves the understanding of information. The development of informational videos requires educational and co-design guidelines to be followed to ensure the standard of the video is reliable and meets consumer needs.

Strengths and limitations

A strength of this research was that interviews were conducted with participants who had various complex cancers, providing a deep understanding of their information needs. Another strength was using Teams for interviews, allowing the interviewer to respond to visual cues and clarify information. There were no major limitations, and this qualitative project deepened our understanding of information needs and preferences for information presentation.

Declarations

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Author Contribution

All authors contributed to the study conception and design. Material preparation, data collection [Elisabeth Coyne] and analysis were performed by all authors. The first draft of the manuscript was written by [Elisabeth Coyne] and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript."

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Figures

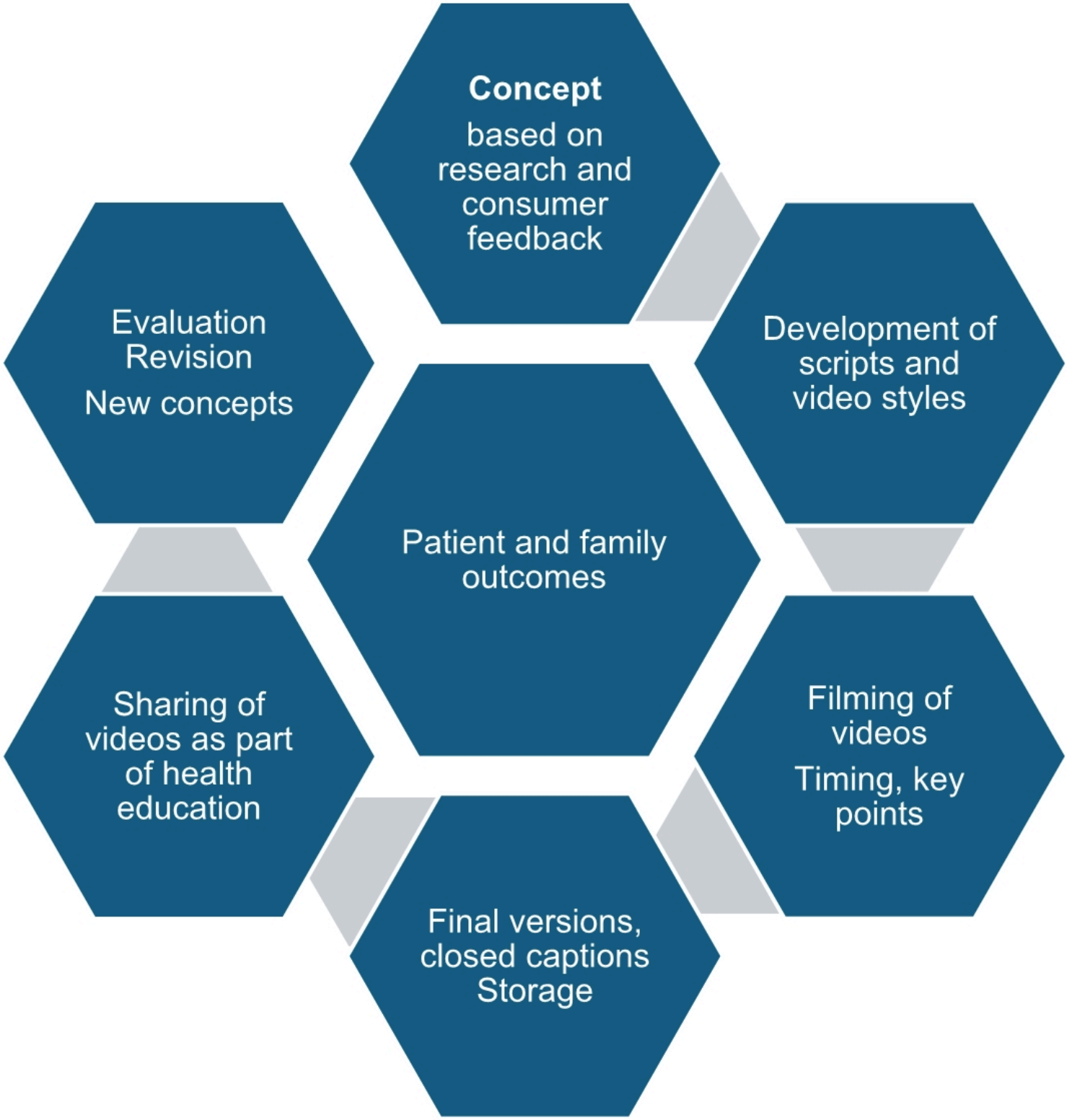


Figure 1

Cyclic Co-design Framework in a study developing videos for complex cancer

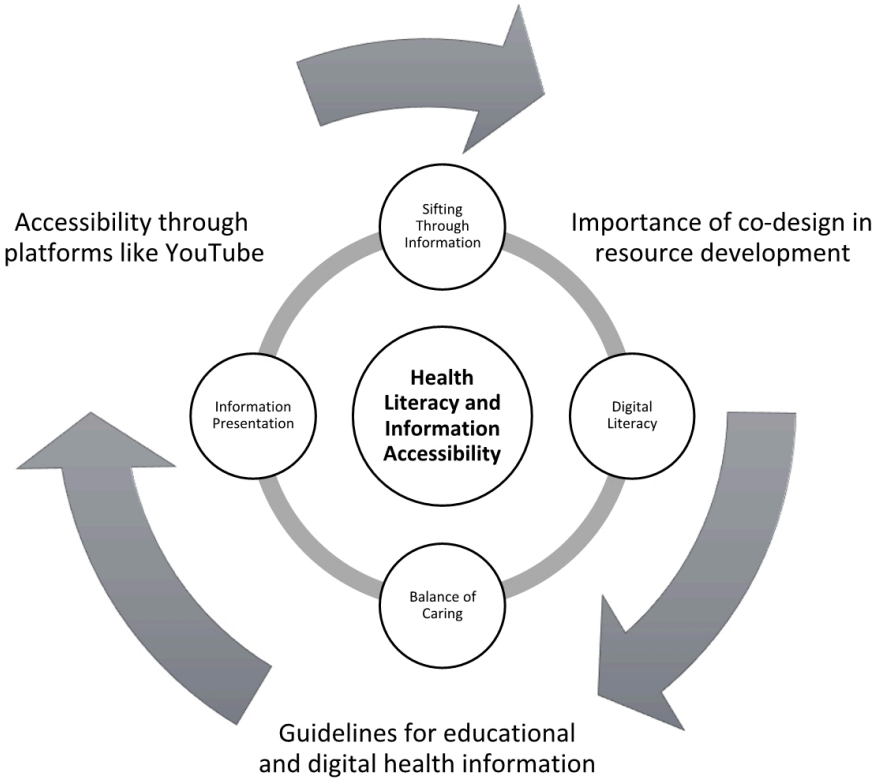


Figure 2

Health Literacy and Information Accessibility Framework for individuals and their families with complex cancer.